

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Third Semester B.E. (Civil) (Credit System) Degree Examinations

April 2023

21CV302 – FLUID MECHANICS AND HYDRAULICS

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

Unit – I

Marks BT* CO*

1. a) Distinguish between:
 - i) Ideal fluids and real fluids ii) Newtonian and non-Newtonian fluids. 5 L*2 1
 - b) A body of mass 100kg slides down at a uniform velocity of 1m/s along the lubricated inclined plane making 30° with the horizontal. The viscosity of the lubricant is 0.1 Pascal-sec and the contact area is 0.25 m². Determine the thickness of the lubricant. 5 L3 1
 - c) Illustrate the difference between absolute pressure and gauge pressure. Convert a vacuum pressure of 5m of water into absolute pressure in terms of N/m². Take atmospheric pressure as 10.33 m of water. 5 L2 2
 - d) A differential monometer is connected at the two ends of a pipe. The pipe A contains liquid of specific gravity 1.5 while pipe B contains liquid of specific gravity 0.9. The pressure at A is 100kN/m² and at B is 180kN/m². The center line of pipe B is 3m below the centerline of A. If the level of mercury in the left limb is 5m below the centerline of A, determine the difference in level of mercury in differential monometer. 5 L2 2
2. a) Illustrate the difference between:
 - i) kinematic viscosity and dynamic viscosity 5 L2 1
 - ii) surface tension and capillarity.
 - b) How the vapour pressure and compressibility of a fluid are defined? A liquid at 20°C has a relative density of 0.8 and kinematic viscosity of 2.3 centistoke. Compute its (i) unit weight, ii) dynamic viscosity. Assume ρ (water) = 998 kg/m³. 5 L2 1
 - c) Distinguish between
 - i) uniform flow and steady flow ii) streamline and stream tube. 4 L2 2
 - d) State and explain Pascal's law. Evaluate the pressure difference between the points A and B in the set-up shown in Fig. 2(d).

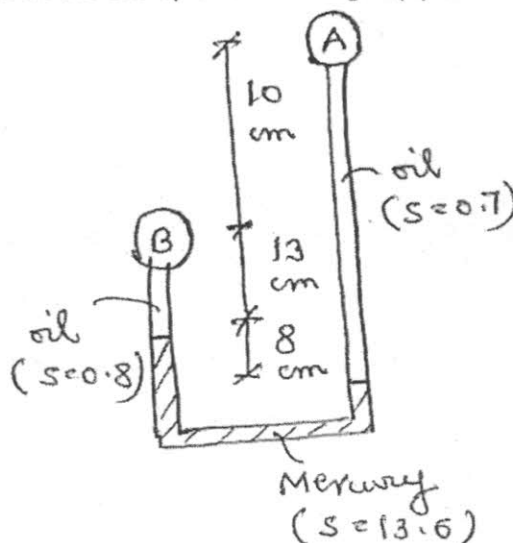


Fig. 2(d)

3. a) State and explain Newton's law of viscosity. What are its limitations? 4 L2 1
- b) A sleeve 10cm long encases a vertical metal rod 3 cm in diameter with a radial clearance of 0.02mm. If it is immersed in an oil of viscosity 6 poise, the effective weight of sleeve is 7.5N, compute the velocity of the rod with which it slides down. 6 L3 1
- c) State and prove hydrostatic pressure law. 5 L2 2
- d) Illustrate the principle of deriving the continuity equation of flow. An incompressible fluid flows steadily through two pipes of diameters 15 cm and 20 cm, which combine to discharge in a 30 cm diameter pipe. If the average velocities in the two pipes are 2 m/s and 3 m/s respectively, compute the velocity in 30 cm diameter pipe. 5 L3 2

Unit - II

4. a) Starting from Euler's equation of fluid motion along a streamline, arrive at the Bernoulli's equation. Highlight the limitations of the equation. 5 L2 3
- b)

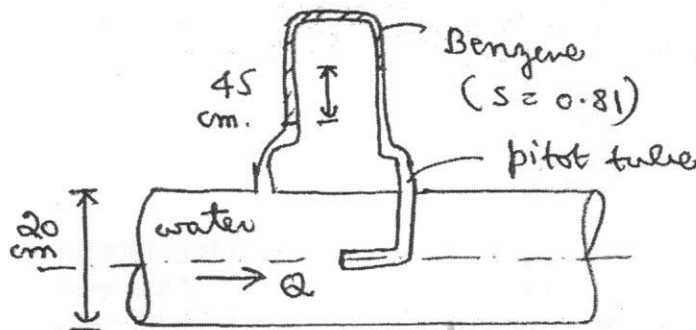


Fig. 4(b)

- A pitot tube is connected to a pipe carrying water as shown in Fig. 4(b). If the mean velocity is 80% of maximum velocity, compute the discharge in the pipe. Assume coefficient of tube as 0.98. 5 L3 3
- c) Define the hydraulic coefficients of an orifice. Arrive at the relationship between them. How an orifice differs from a mouthpiece? 6 L2 4
- d) Water flows over a triangular right angled weir and then over a rectangular weir of 1m crest length. The discharge coefficients are 0.6 and 0.7 respectively. If the depth of water over triangular weir is 360mm, compute the depth of water over the rectangular weir. 4 L3 4
5. a) A pipeline 320m long is laid at an angle of 6° with the horizontal. It tapers from 1.25m diameter at the higher end to 0.70m diameter at the lower end. The pipe carries gasoline of specific gravity 0.815. The pressure at higher end and lower end are 70kPa and 74kPa respectively, compute the loss of head, if the discharge in the pipe is $1.25\text{ m}^3/\text{s}$. 4 L3 3
- b) The velocity of water in a 60cm ϕ and 15mm thick cast iron pipe ($E=1.04 \times 10^{11} \text{ N/m}^2$) is changed from 3m/s to zero in 1.25 seconds by closure of a valve. If the pipe length is 800m, compute the water hammer pressure at the valve. Evaluate the pressure rise if the closure takes place in 3 seconds. Assume K for water as $2.11 \times 10^9 \text{ N/m}^2$. 6 L3 4
- c) Derive the discharge equation for a V-notch. 5 L2 4
- d) A reservoir discharges through an orifice (sluice) 915 mm wide and 1,220 mm deep. The top of the sluice is 610mm below the water level in the reservoir and the downstream water level is below the bottom of the sluice. Estimate i) the discharge in the sluice taking $C_2 = 0.6$, ii) % error in computing the discharge if the sluice is treated as a small orifice. 5 L3 4
6. a) List the different losses that encounter in a pipe flow. Illustrate with sketches and relevant expressions. 5 L1 3

- b) A venturimeter of throat diameter 50mm is fitted into a 125mm diameter water pipeline. Compute the flow in the pipeline if the reading of a mercury-water differential U-tube manometer connected to upstream and throat sections is 200mm. Take $C_d = 0.96$. Sketch the arrangement.
- c) Classify orifice based on various criteria. Illustrate with sketches.
- d) List the salient features of a Cippolletti notch. Estimate the discharge through a trapezoidal notch 1.2m wide at top and 0.5m at the bottom. The head over the notch is 0.3m. Take C_d for triangular portion as 0.60 and for rectangular portion as 0.62.

5	L3	4
5	L1	4
5	L3	4

Unit – III

7. a) Compare and contrast:
i) Gradually varied flow and suddenly varied flow.
ii) pipe flow and open channel flow.
- b) A trapezoidal channel with side-slopes of 2H:1V has to be designed to carry $15\text{m}^3/\text{s}$ at a slope of 1:5000. Compute the dimensions of the efficient section. Take Manning's $n = 0.014$.
- c) Deduce that the maximum efficiency of a jet striking a series of flat vanes mounted on a wheel is 50%.
- d) A jet of water of diameter 50mm moving with a velocity of 25m/s impinges on a fixed curved vane tangentially at one end at an angle of 30° to the horizontal. Calculate the resultant force of jet on the vane, if the jet is deflected by 50° .
8. a) Define uniform flow in open channels. A trapezoidal channel has a bed width of 2.0 m and side slopes of 1.5 H:1V. The channel has a longitudinal slope of $\left(\frac{1}{4000}\right)$. If the Manning's $n = 0.018$, compute the discharge in the channel when the depth of flow is 1.4m.
- b) Define most economical channel section. Deduce that the hydraulic radius is half the depth of flow for an efficient trapezoidal channel section.
- c) State and explain Impulse-Momentum Principle. Give the expressions for the force exerted by a jet on (i) a stationary inclined flat vane (ii) a moving symmetrical curved vane (striking at centre).
- d) A jet of water of 30mm ϕ strikes a hinged square plate (hinged at top) at its centre with a velocity of 20 m/s. The plate is deflected through an angle of 20° . Compute the weight of the plate.

5	L2	5
5	L3	5
5	L2	5
5	L3	5
5	L3	5
5	L2	5
5	L2	5
5	L3	5

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome
